Psychology of Learning: Who Gave What

NPTEL · IIT Kharagpur · Prof. Atasi Mohanty. Every person named in the weekly assignments from 2023 to 2026, with the theory the lecture credits them for, in course order.

How to read this. Bold words are the keywords the answer options use. The red number is how many assignment questions (all years) ask about that person, so learn the high numbers first. Where the lecture differs from textbooks (Murray and arousal theory, for example), this sheet follows the lecture, because NPTEL marks by the lecture.

Weeks 1–2 · Foundations and major theories of learning

Person	What they gave	Asked
Wilhelm Wundt 1879	**Father of modern psychology.** Set up the **first psychology laboratory** at the University of **Leipzig, Germany** (1879).	4×
John B. Watson behaviorism	**Founder of behaviorism.** Famous work: **"Psychology as the Behaviorist Views It"**. Saw psychology as an **experimental** science of observable, measurable behaviour. **Little Albert** experiment: conditioned a child to **fear** a white rat. First described the basic stimulus–response model.	5×
Ivan Pavlov behaviorism	**Classical conditioning** (bell + food → salivation). Pioneer of behaviorism with Watson and Skinner.	1×
B. F. Skinner behaviorism	**Operant conditioning**: behaviour is strengthened or weakened by its consequences (reinforcement, punishment). Built on Watson's S–R model.	–
Tolman, Bruner and the German Gestalt psychologists (Köhler, Wertheimer)	Their work was **instrumental in the shift from behaviorism to cognitive theories**. If a question lists several of them, the answer is usually **"All of the given"**. Gestalt psychologists criticised behaviorism.	1×
Albert Bandura	**Observational learning** (attention, retention, reproduction, motivation). Week 11 covers his **social cognitive theory** in detail.	4×
Howard Gardner	**Theory of multiple intelligences. Intrapersonal** = self-knowledge and acting adaptively on it; **interpersonal** = understanding others; **spatial** = perceiving the visual-spatial world accurately (colour, line, shape, form, space).	2×
Robert Sternberg 1985	**Triarchic theory of intelligence**, proposed as an **alternative to the general intelligence factor (g)**. He argued g measures only **academic** intelligence.	1×
Charles Spearman	**General intelligence factor, "g"**: what IQ tests typically measure. *Textbook; the lecture discusses g without naming him.*	2×
J. P. Guilford 1955	**Structure of Intellect (SOI)** model: three dimensions (contents × operations × products) giving **180** abilities. The products are units, classes, relations, systems, transformations, implications. "Symbolic" is a content, **not** a product.	1×
J. P. Das (with Naglieri)	**PASS theory** of intelligence: **P**lanning, **A**ttention-arousal, **S**imultaneous and **S**uccessive processing.	2×
Binet and Simon	**Binet–Simon scale**: the first intelligence (IQ) test, which started the era of measuring intelligence.	–
Jean Piaget	**Cognitive constructivism** and cognitive development in **4 stages**: sensorimotor, pre-operational, concrete operational, formal operational ("post-operational" is **not** a stage). Teaching should suit the learner's developmental stage. Week 11: **cognitive conflict** triggers assimilation and accommodation.	2×

Person	What they gave	Asked
Lev Vygotsky	**Social constructivism**, which adds the influence of society and peers on learning. **Zone of Proximal Development** = the gap between what a learner can do alone and with help from a more knowledgeable other.	3×
Ernst von Glasersfeld	**Radical constructivism**. The three types of constructivism are cognitive (Piaget), social (Vygotsky) and radical; "biological constructivism" is not one.	–
Robert Gagné	**Conditions of Learning** = **Nine Events of Instruction**, a framework for designing instruction. His definition of learning is in Week 7.	4×

Week 3 · Memory and cognition

Person	What they gave	Asked
Atkinson and Shiffrin	**Multi-store memory model** (information-processing approach): sensory register → short-term memory → long-term memory. The **sensory register** has a very large capacity but lasts only about 1–3 seconds.	2×
Alan Baddeley (and Hitch)	**Multicomponent model of working memory**: central executive, phonological loop, visuo-spatial sketchpad, episodic buffer. "Language repository" is **not** a component.	2×
Craik and Lockhart	**Levels of processing** theory: **shallow** processing encodes only physical qualities, **deep** processing encodes meaning.	1×

Week 4 · Thinking and cognition

Person	What they gave	Asked
Benjamin S. Bloom 1956	**Bloom's taxonomy**: hierarchical models that classify **learning objectives** by complexity. **Three domains**: cognitive, affective, **psychomotor** (physical or skilled behaviour). The lowest level is **Remember**. "Biological" is not a domain.	6×
Anderson and Krathwohl 2001	**Revised Bloom's taxonomy**: Remember → Understand → Apply → Analyse → Evaluate → Create.	–

Week 5 · Motivation

Person	What they gave	Asked
Clark Hull 1943	**Drive (drive-reduction) theory**. A behaviourist who proposed **three primary drives** that drive human behaviour (biological drives such as food and water).	2×
Henry Murray 1938	The lecture slide says the **Arousal Theory of Motivation was first proposed by Henry Murray in 1938**. Answer "Murray" even though textbooks link arousal to Yerkes–Dodson.	1×
Yerkes and Dodson	**Yerkes–Dodson law**: an **inverted-U** relationship between **arousal** and **performance**. Performance improves up to an optimum, then falls off.	2×
Victor H. Vroom 1964	**Expectancy theory**: Expectancy × Instrumentality × **Valence**. Valence is the perceived value of the reward; it is **negative** when a person wants to avoid that reward.	5×
Abraham Maslow	**Hierarchy of needs, 5 tiers** (bottom to top): **physiological** (food, water, warmth, sleep) → safety → **love and belonging** (friendship) → **esteem** (prestige, accomplishment) → self-actualization. "Safety needs at the bottom" is **false**.	5×

Person	What they gave	Asked
Deci and Ryan 2000	**Self-determination theory**: the 3 psychological needs are **autonomy, competence, relatedness**.	5×
Edwin Locke	**Goal-setting theory**: set clear, usable targets for learning and performance.	2×

Week 6 · Emotional intelligence

Person	What they gave	Asked
Peter Salovey and John Mayer 1990	**Proposed** the theory of emotional intelligence (ability model): EQ is a **set of skills**.	–
Daniel Goleman 1995	**Popularised** the term "emotional intelligence" in his 1995 book. **Five areas**: self-awareness, self-regulation, motivation, **empathy**, social skills. Empathy and social skills cover **handling relationships with others**. "Knowledge" is **not** a component.	4×
Reuven Bar-On	**Mixed model**: EQ is a combination of competencies, skills and **"facilitators"**. **10 distinct components** scaffold emotionally and socially intelligent behaviour. Facilitators: self-actualization, independence, social responsibility, optimism, happiness.	2×
Arlie Hochschild	**Emotional labour**, using the example of **flight attendants**. **Feeling rules** = organisations' attempts to control employees' inner thoughts and feelings.	3×

Week 7 · Learning and instruction

Person	What they gave	Asked
Herbert Simon	Learning = **changes in a system that result in improved performance** over time on tasks similar to those done before.	1×
Robert Gagné	Learning = "a **change in human disposition or capacity** that **persists** over a period of time and is **not simply ascribable to processes of growth**." The false option is the one saying it *is* due to growth.	4×
John Dewey	**"Reflection, learning and education"**. **Returning to experience** = recalling or detailing salient events.	2×
Carl Rogers	**Humanistic orientation** to learning: personal involvement of the **whole person**, both feeling and cognitive.	5×
David Ausubel	**Meaningful learning** happens when new information is integrated into existing knowledge structures (mental schemas).	2×

Week 8 · Learning and pedagogy

Person	What they gave	Asked
David Kolb	**Experiential learning, a 4-stage cycle**: concrete experience → reflective observation → abstract conceptualisation → **active experimentation (planning / trying out)**. The cycle sits on **two continuums: perception and processing**. Styles: **diverging** (watch rather than do, multiple perspectives), assimilating, **converging** (technical tasks, likes to be left alone), accommodating.	8×
Richard Mayer	**Multimedia learning**, which rests on three assumptions: two separate channels (visual and auditory), each with limited capacity, and active processing.	3×
Jack Mezirow 1978	**Transformative learning**: **10 steps**, starting with a **disorienting dilemma**.	2×

Week 9 · Psychology of e-learning

Person	What they gave	Asked
Malcolm Knowles 1913–1997	**Andragogy** (adult learning): **5 assumptions** (self-concept, experience, readiness, orientation, motivation to learn) and **4 principles** (for example, adults must have a hand in designing their own learning).	–
David Merrill 2002	**Principles of Instruction**, **5 principles**: task-centred, activation, demonstration, application, integration.	1×
Gagné / Bloom (again)	Week 9 slides reuse **Gagné's Nine Events** and the **revised Bloom's taxonomy** (Anderson and Krathwohl, 2001) for e-learning design.	–

Weeks 11–12 · Social cognition and sustainability

Person	What they gave	Asked
Albert Bandura	**Social cognitive theory**: observational learning, modelling, **self-efficacy**. Central tenet: people seek a sense of **agency**. Three models of observational learning: **live** (a real person demonstrating), symbolic, verbal instructional.	4×
Norman Anderson 1981	**Information integration theory**: impressions are a weighted average of the perceiver's dispositions and the target's traits.	–
Eggert 2009	Sustainability, as used in environmental, economic and social sustainability, is **unidimensional** (one-dimensional). **Sustainable development** is **multidimensional**.	3×

Numbers and years to remember

Maslow's hierarchy	**5 tiers**	Knowles: assumptions / principles	**5 / 4**
Kolb's learning cycle	**4 stages**	Merrill's principles	**5 (2002)**
Piaget's development stages	**4**	Wundt's lab, Leipzig	**1879**
Goleman's EI areas	**5**	Murray, arousal theory	**1938**
Bar-On's components	**10**	Hull, drive theory	**1943**
Mezirow's transformative steps	**10 (1978)**	Vroom, expectancy theory	**1964**
Gagné's events of instruction	**9**	Sternberg, triarchic theory	**1985**
Guilford's SOI abilities	**180 (1955)**	Salovey and Mayer, EI	**1990**
Bloom's domains	**3 (1956)**	Goleman's book	**1995**
Self-determination needs	**3**	Revised Bloom (Anderson and Krathwohl)	**2001**

Names that are easy to mix up

- **Popularised EI** = Goleman (1995). **Proposed EI** = Salovey and Mayer (1990). **Mixed model, 10 components** = Bar-On.
- **Drive theory / 3 primary drives** = Hull. **Arousal theory** = Murray (per the lecture). **Inverted-U** arousal–performance = Yerkes–Dodson.
- **Cognitive constructivism, stages** = Piaget. **Social constructivism, ZPD** = Vygotsky.
- **Multiple intelligences** = Gardner. **Triarchic, alternative to g** = Sternberg. **g factor** = Spearman. **180 abilities, SOI** = Guilford. **PASS** = Das.
- **Working memory (multicomponent)** = Baddeley. **Multi-store (sensory → STM → LTM)** = Atkinson and Shiffrin. **Levels of processing** = Craik and Lockhart.
- **Reflection, "returning to experience"** = Dewey. **Experiential cycle** = Kolb. **Transformative, 10 steps** = Mezirow. **Humanistic** = Rogers. **Andragogy** = Knowles. **Meaningful learning** = Ausubel.
- **Classical conditioning** = Pavlov. **Operant conditioning** = Skinner. **Little Albert, "Psychology as the Behaviorist Views It"** = Watson.
- **Freud** (psychoanalysis) and **Jung** (analytical psychology) are **not** behaviorists. They appear only as wrong options. **Wundt** = first lab and father of modern psychology.
- **Learning definitions**: "improved performance on similar tasks" = Simon. "Change in disposition or capacity… not ascribable to growth" = Gagné.